

EMSOFT 2026 Artifact Submission Overview

Paper: “Suborbital and Orbital Real-Time Localization of Gamma-Ray Bursts via Utility-Guided Coordination”

The corresponding information for this artifact is provided below.

1. Artifact Access and Archival Repositories

- **Source-code and instructions repository (includes README, REQUIREMENTS, etc.):**
<https://github.com/Daisy0419/GRB-Search-Pipeline.git>
- **Software artifact:**
<https://zenodo.org/records/21572120> DOI 10.5281/zenodo.21572120
- **Artifact dataset:** <https://zenodo.org/records/21497891> DOI 10.5281/zenodo.21497891

Note: Please visit the **GitHub repo** above as the initial step, and its **README.md** file in particular.

The existing Zenodo dataset contains the large detector models and simulated transient datasets required by the experiments. The version-specific Zenodo artifact snapshot, containing the source code, artifact instructions, environment specifications, precomputed results, analysis notebook, accepted paper version, and open-source license, is currently being prepared.

The authors will inform the Artifact Evaluation Committee and update the HotCRP submission as soon as the corresponding version-specific DOI is available. Continued development may take place in the GitHub repository, but the forthcoming Zenodo DOI will identify the exact GitHub repo version submitted for evaluation.

2. Expected Technical Skills and Execution Environments

Expected reviewer skills

Reviewers should be comfortable with:

- using a Linux command-line environment;
- creating and activating Conda environments;
- building C++ software with CMake and Make;
- configuring environment variables;
- running Python scripts; and
- using Jupyter Notebook to regenerate and inspect the paper figures.

No specialized astronomy knowledge is required to install or execute the artifact.

Operating systems and processor architectures

- **Complete training and evaluation workflow (README Sections 1–3):**
Linux on ARM64 (`aarch64`).
- **Optional Intel benchmark (README Section 4):**

Linux on Intel/AMD x86-64.

The reference ARM platform used for the paper is an NVIDIA Jetson Orin NX with:

- eight ARM Cortex-A78AE v8.2 CPU cores;
- CPU frequency fixed at 1.5 GHz; and
- 16 GB DRAM.

The GPU is not used. No other non-commodity peripheral is required.

The reference Intel platform used for the Intel measurements in Figure 4 is a 2.3 GHz Intel Xeon Gold 5118 system with 256 GB DRAM.

The reported execution times are platform-sensitive. Functional execution on another compatible platform does not imply that its timing measurements will match those reported in the paper. Exact timing comparisons require the corresponding reference platform.

Required software

The artifact requires:

- Git;
- Bash and standard Linux command-line tools;
- `wget` or `curl`, `tar`, and `md5sum`;
- Conda or Miniconda;
- Python and the packages specified by:
 - `cosipy-312-deps.yaml` for ARM64; and
 - `cosipy-312-intel.yaml` for the optional Intel benchmark;
- Jupyter Notebook;
- CMake and Make;
- a C++17 compiler with OpenMP support;
- LEMON Graph Library 1.3.1;
- HDF5 with its C++ and high-level libraries enabled;
- HighFive headers;
- the Bitshuffle HDF5 plugin supplied by the Python `hdf5plugin` package; and
- the `tsmap-artifact` branch of COSIpy, installed according to `INSTALL.md` and `README.md`.

No licensed or proprietary software is required.

Storage, data, and network requirements

- Source repository: approximately 200 MB.
- Miniconda installation: approximately 800 MB.

- Conda environment: approximately 2.8 GB.
- LEMON installation: approximately 70 MB.
- Compressed dataset download: approximately 16.2 GB.
- Extracted detector models and transient datasets: approximately 25–30 GB.
- Maps generated by the complete experiments: approximately 30 GB.
- Required storage after extraction: at least 65 GB, plus temporary space for the downloaded archive.

Network access is required during installation to obtain the repositories, software dependencies, and Zenodo dataset. After installation and data download, the experiments can be executed offline.

Evaluation paths and approximate runtimes

The artifact provides three evaluation paths:

1. **Reproduce Figures 4–11 from the paper results.**
This is the shortest evaluation path and should take minutes.
2. **Regenerate the complete training and evaluation results.**
The training phase takes approximately 18 hours, and the evaluation phase takes approximately 21 hours on the reference platform.
3. **Optionally rerun the platform-sensitive mapping benchmark for Figure 4.**
The Intel measurements take approximately 20 minutes, and the Jetson measurements take approximately 10 minutes.

Figures 1–3 are explanatory diagrams and are not generated by the artifact.

Detailed installation, smoke-test, execution, and expected-output instructions are provided in `INSTALL.md` and `README.md`.

3. Requested Artifact Badges

The authors apply for all three EMSOFT 2026 artifact badges:

- **Available**
- **Reviewed**
- **Reproducible**

The source code, documentation, environment specifications, precomputed results, analysis workflow, and accepted paper are provided through the GitHub repository. The large detector models and simulated transient datasets are persistently available through the dataset DOI above.

The version-specific Zenodo snapshot of the source code and artifact instructions is currently under preparation. The authors will inform the Artifact Evaluation Committee and update the submission as soon as its DOI is available.

The accepted paper version is available on the following pages